

Language of gleam: Impressionism Artwork Automatic Caption Generation for People with Visual Impairments

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ABSTRACT

User Experience Design (UX Design) comes from focusing on how products, in reality, affect the user's experience. In particular, the design of multi-modal interfaces for blind people facilitate the flexible and natural product or service capacity and improve the interaction of blind people by allowing them to overcome the various existing constraints associated with any particular interaction. There have been various attempts to help visually impaired people appreciation of visual artwork, including multi-modal associations. However, these methods can only provide general information in terms of edge and pattern recognition by the sense of touch and are restrained by availability and number of specially developed artworks. We propose a novel method explaining visual artworks through image caption generation using artificial intelligence (AI) to improve artwork accessibility. This method can objectively describe any impressionism artwork that might be used as a standalone description of art interpretation for blind people or can aide tactile-based methods. Based on end-to-end learning with a deep neural network, an encoder-decoder architecture model is adopted, and comprehensive experiments are performed to evaluate the stability of generated image captioning for stylized COCO datasets with impressionism.

Keywords: visual artwork, image caption generation, human-centric artificial intelligence, deep learning,

1. INTRODUCTION

Assistive Technology (AT) is usually referred to as auxiliary, adaptive, and rehabilitation devices that can alleviate inconveniences in everyday life due to mental or physical disabilities or deterioration of physical function due to aging [1]. The social impact of auxiliary science and technology contributes to the exercise of basic rights for impaired users to equality, dignity, and respect [2]. It has historically made a significant contribution to the lives of the disabled and the elderly, and the need is becoming more prominent as the number of registered disabled people continues to increase and enter the aging society. In line with the historical background of Information and Communications Technologies (ICT), the museum will expand its appreciation of the area of art beyond simple print or computer-based information delivery to reduce the digital divide in exhibition viewing. Finally, the user experience design derived from human engineering suggests that it has excellent ease of applying to auxiliary science and technology. As shown in Figure 1, It intends to establish a non-visual information delivery interface to realize the expansion of the right to enjoy a blind culture.

Metaphor in User Interface Design (UI Design) aims to provide familiarity with the product's use, thereby improving the user's immediate model understanding and accessibility. It means that users can quickly identify a given system by utilizing the knowledge category as an inherent lowest common denominator. Early-stage prototyping designs are becoming increasingly important, and prototyping methods and fidelity settings ensure conditions that enable efficient and sustainable product development in terms of time and cost [3]. Therefore, with the minimum heuristics and low fidelity prototype types, the focus was on providing a service to listen to visual artworks' commentary or explanation by utilizing hearing, rather than viewing through contrived tactile sensations. The purpose of the module, which consists of two parts: image caption generation and text to speech, was to promote the cultural enjoyment of the blind and the development of exhibition art appreciation tools. In this paper, we design image caption generation, which is the pre-stage of the final module so that visually impaired people can appreciate art without relying on direct help from art critics.

Around 217 million people around the world are people with visually impaired (PVI). These people, especially with total blindness, are unable to experience visual art. Even though their sense of vision is restricted, the information may still

be conveyed through other senses to some extent. The right of disabled people to participate in all aspects is agreed upon by the United Nations. Multiple schemes for PVI's art interaction have been proposed utilizing art-tactile cross-modal association as PVI has superior performance over tactile tasks [11,12]. A 3D printed 2.5D model has been proposed in [13]. However, availability and variety of artwork tactile and 2.5D models to PVI's are minimal. Also, the PVI's have to navigate their finger all through the tactile pattern in order to interpret the boundary of pattern into an object from artwork, and this problem intensifies for the case of overlapping and non-tangible objects from artwork for identification issues. Also, the color, depth, and artwork emotion aspects are not conveyed through tactile and 2.5D PVI touch perception. Audio guides are a standard method to provide art description to PVI verbally [14], but they also are subject to availability. There is a need for the development of a system that can assist PVI while browsing through artwork in order for them to choose any artwork for exploration in detail using existing methods. The method should also cater to the need of PVI to get information for any artwork without the need for separate audio guide preparation and should be able to respond to questions so that most of the art information can be conveyed to PVI's.

We have proposed a deep learning-based scheme that can automatically generate a short description of the given artwork. This scheme can be extended for verbal communication and question-answer based system for PVI's. An encoder-decoder based attention-on-attention model has been utilized for artwork caption generation. In order to evaluate the deep learning model, artwork imitated images are fed to deep learning models, and performance is measured with related evaluation metrics. The imitated artwork images are stylized using neural style transfer for impressionism art genre. The proposed scheme has been presented in Fig. 1. The proposed scheme can be extended for other artwork genres and detailed description generation for question-answer based model.

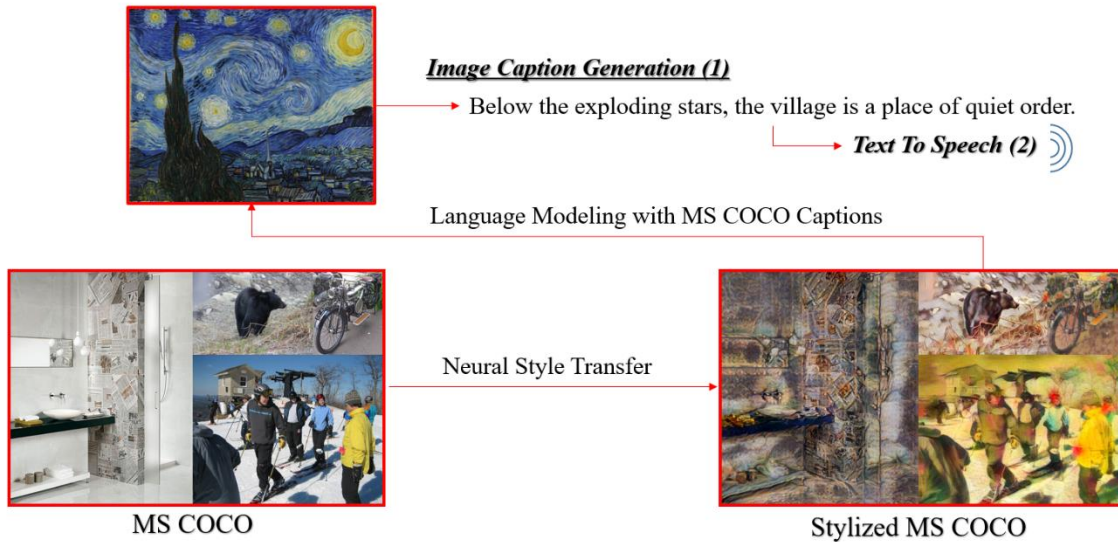


Figure 1. Language of gleam: Scheme for auditory feedback on artwork for people with visual impairments

2. PRELIMINARIES

2.1 Deep Learning Based Image Captioning

Several kinds of literature usually refer slightly differently to a little bit of the details of image captioning. However, there is a typical workflow in which a computer recognizes and expresses a given image, and it approaches image caption generation in three main ways, as shown following. The based-retrieval method is a method to select the captions for the query image from these captions pools after candidate captions are defined [4]. However, they have weakness fault that they cannot generate images semantically correct captions and becomes difficult to analyze the image correctly outside the scope of the training data set. The task of captioning images is mostly divided into two parts: look and language expression. This could make it difficult to manage tasks efficiently. After templates with some blank slots, the based-template method requires to fill the slots for generating image captions [4]. It produces grammatically accurate captions, but it has the disadvantage of not expecting colorful expressions that deviate from the form of a given template. Based-end-to-end

learning with deep neural network means a deep learning-based methodology that can form feature maps of images to generate captions in multimodal space [4]. Forming feature maps in a multimodal space beyond the existing visual space induces, creating vivid expressions. Most image caption generation-related research is being conducted focusing on attention, semantic concept, novel object, and styled caption. Approach based end-to-end learning is the latest method for research area about image captioning, usually carried out by adopting an encoder-decoder architecture. A learning way for image caption generation is a supervised learning text-sequences corresponding to images.

Artificial intelligence has ceaselessly challenged a wide selection of problems, such as knowledge discovery and pattern recognition. Image captioning could be particularly deemed a re-representation for a given image from the standpoint of cognitive engineering. In other words, the field of image captioning plays a significant role as a bridge between computer vision and natural language processing (NLP) [5]. For example, Image schemas for re-representation might require three types of operation, which include the concept of "merge", "collection", "structured" cognitively-based ontology design patterns [6]. Authors represented the first form of image schematic combination that atomic combinations turn into complex image schemas and the second form as classical image schema profiles and the last form as sequential image schema combinations [7]. However, the concept of ontology in computer science is close to concentrating on computability for being than the metaphysical meaning of being.

Attention mechanism on the field of natural language processing and computer vision can be corresponding to process capturing a kind of abstract domain ontologies. It generates captions about visual artwork images, applying to stylized MS-COCO dataset which origin MS-COCO dataset [8] is converted by neural style transfer technique (Adaptive Instance Normalization, AdaIN) into [9]. The attention mechanism has been considered as a productive potential to improve the performance of language models. Along with these tendencies, several variants of the transformer have been proposed lately and have presented meaningful results. [10] Based on statistical learning theory, image captioning with a deep neural network has kept being developed over some years as a representative example of a one-many structure. Conspicuously, it has a vast analogy with a neural machine translation (NMT) like a structural feature of encoder-decoder architecture generating latent vectors. It can be useful to gain a fair approach to the plausible insight for end-to-end learning on image captioning by grasping NMT up to the hilt. To be specific, sequence to sequence language model and attention mechanism are crucial factors figuring out applications. However, there exist still the most prominent technical issue of objective appraisal metric and overfitting in image captioning. Because the overfitting issue mainly creates ambiguous or stereotypical descriptions for an image, it remains a challenge to be solved.

2.2 Dataset Description

Publicly available Microsoft COCO: common objects in context (COCO) dataset has been utilized for experimentation in this study [7]. It is the most significant benchmark dataset containing 123,287 images labeled with five captions for each image. The dataset has been divided into 82,783 training, 40,504 validation, and 40,775 tests set images, applying 'Karpathy' data split. All of these images contain photos of multiple real-world objects from 91 object types. As deep neural network-based AI systems need an unprecedented number of labeled training samples, and no aforesaid dataset for artwork is available. Therefore, the transformed version of the original dataset, hereby referred to as stylized MS-COCO, has been utilized to evaluate our proposed model, which is elaborated in the subsequent section.

2.3 Neural Style Transfer: Adaptive Instance Normalization (AdaIN)

In style transfer, the AdaIN overcoming slow iterative optimization process has a strength of flexible user controls by adjusting the style weight on loss function [19]. It is advanced normalization technique computing affine parameters from the style image, so that has an effect on style swapping without computation cost. The encoder-decoder architecture consists of the first layers up to 'relu4_1' of a pre-trained VGG 19 and its mirrors with all pooling layers replaced by nearest up-sampling without normalization layers. The AdaIN layers that produce target feature maps or latent vector in feature space are as follows,

$$AdaIN(x, y) = \sigma(y) \left(\frac{x - \mu(x)}{\sigma(x)} \right) + \mu(y) \quad (1)$$

$$x = f(c), y = f(s) \quad (2)$$

where $\mu, \sigma \in \mathbb{R}^{N \times C}$ are mean and standard deviation; f, c, s are encoder and content, style images. If target feature map put t , that is AdaIN output, loss for training decoder g are defined as follows,

$$\mathcal{L} = \mathcal{L}_c + \lambda \mathcal{L}_s \quad (3)$$

$$\mathcal{L}_c = \|f(g(t)) - t\|_2 \quad (4)$$

$$\mathcal{L}_s = \sum_{i=1}^L \|\mu(\phi_i(g(t))) - \mu(\phi_i(s))\|_2 + \sum_{i=1}^L \|\sigma(\phi_i(g(t))) - \sigma(\phi_i(s))\|_2 \quad (5)$$

where each ϕ_i denotes feature map of a layer in VGG-19. Style interpolation for arbitrary style transfer is achieved by construction of target feature map to being as follows with a affine set of K style images feature and corresponding weights ($\sum_{k=1}^K w_k = 1$) for local preservation of convexity within the limits of the possible [15].

$$t = g\left(\sum_{k=1}^K w_k \text{AdaIN}(f(c), f(s_k))\right) \quad (6)$$

2.4 Image Captioning: Attention on Attention (AoA)

In image captioning, AoA generates attended information from the attention gate and information vector with GLU (Gated Linear Units) [16]. Theoretically, it has a wide selection of generality that can be applied to any context vector in sequence modeling without requiring hidden states like LSTM/GRU cells to raise serial computation time cost. The author of the original paper presented ‘AoA is a general extension to attention mechanisms’ [17]. AoA module generates ‘information vector’ and ‘attention gate’ and obtains attended information as follows,

$$\mathbf{i} = W_q^i \mathbf{q} + W_v^i \hat{\mathbf{v}} + \mathbf{b}^i \quad (7)$$

$$\mathbf{g} = \sigma(W_q^g \mathbf{q} + W_v^g \hat{\mathbf{v}} + \mathbf{b}^g) \quad (8)$$

$$\text{AoA} : \hat{\mathbf{i}} = \mathbf{g} \odot \mathbf{i} \quad (9)$$

where $W_q^i, W_v^i, W_q^g, W_v^g \in \mathbb{R}^{D \times D}$, $\mathbf{b}^i, \mathbf{b}^g \in \mathbb{R}^D$ and D is the dimension of $\mathbf{q}, \mathbf{v}$, which are features extracted from CNN (Convolutional Neural Network), that is decoder feature above style transfer. And $\sigma, \odot$ denote the sigmoid activation function and the hadamard product. $\hat{\mathbf{v}}$ is the attention result for j-dimension.

$$\hat{v}_i = \sum_j \text{softmax}(f_{\text{similarity}}(\mathbf{q}_i, \mathbf{k}_j)) v_j \quad (10)$$

AoA^E in encoder play a role in refining feature vectors, self-attentive multi-head attention, and the output with residual skip connection is normalized in the layer. The normalized feature with AoA^D in decoder put focus on generation of context vector $\mathbf{c}_t$. The each attention weight, $W^{Q_d}, W^{K_d}, W^{V_d}, W^{V_e}, W^{V_e} \in \mathbb{R}^{D \times D}$ has 8 heads. Hidden state and cell state $\mathbf{h}_t, \mathbf{c}_t \in \mathbb{R}^D$ is maintained by visual vector $\bar{\mathbf{a}} + \mathbf{c}_{t-1}$ and word embedding vector ($W_e \Pi_t$) of $\mathbf{w}_t$. And $W_e \in \mathbb{R}^{E \times |\Sigma|}$ is embedding matrix with the size $|\Sigma|$ of vocabulary and Π_t is one hot vector at time step t.

$$\mathbf{c}_t = \text{AoA}^D(f_{\text{mh-att}}(W^{Q_d}[\mathbf{h}_t], W^{K_d} A^{K,V}, W^{V_d} A^{K,V})) \quad (11)$$

$$A^{K,V} = \text{LayerNorm}(A + \text{AoA}^E(f_{\text{mh-att}}(W^{Q_e} A, W^{K_e} A, W^{V_e} A))) \quad (12)$$

$$\mathbf{h}_t, \mathbf{c}_t = \text{LSTM}(\mathbf{x}_t, \mathbf{h}_{t-1}, \mathbf{c}_{t-1}) \quad (13)$$

$$\mathbf{x}_t = [W_e \Pi_t, \bar{\mathbf{a}} + \mathbf{c}_{t-1}] \quad (14)$$

Language model to generate captions is defined as conditional probabilities on the vocabulary where $W_p \in \mathbb{R}^{D \times |\Sigma|}$ and loss function is set as its negative log likelihood:

$$p(\mathbf{y}_t | \mathbf{y}_{1:t-1}, I) = \text{softmax}(W_p \mathbf{c}_t) \quad (15)$$

$$L_{XE}(\theta) = - \sum_{t=1}^T \log(p_{\theta}(\mathbf{y}_t^* | \mathbf{y}_{1:t-1}^*)) \quad (16)$$

3. METHODOLOGY

The term of impressionism started from strong opposition or criticism for conventional art was provoked the critic Louis Leroy in an exhibition in Paris on April 25, 1874. Impressionism art genre expresses individual freedom and subjectivity by emphasizing brushed colors, intensity, and texture based on the illustration of the realistic scene of modern life and is regarded as the beginning of contemporary or modern art. MS-COCO dataset [7] is closest to impressionism among these different genres of art, so it can be used to provide an indirect description of impressionism painting works after impressionism style transfer. Besides, the impressionism art genre has an overwhelming number of works compared to other art genres in addition to its broad appreciation, making it a perfect candidate for this application of the deep learning-based scheme

Overall workflow, as shown in Figure 2, has two tasks, neural style transfer and caption generation on the outline. Each ‘Adaptive Instance Normalization’ [9] and ‘Attention on Attention’ [17] techniques are selected for neural style transfer and captioning process. The former has transfer property from arbitrary style image onto given content. In contrast, the latter has a broad generality capability by the generation of highly accurate captions and proficiently figure out the interactions of objects in an image through enhanced global context vector, details of which have been provided in sub-

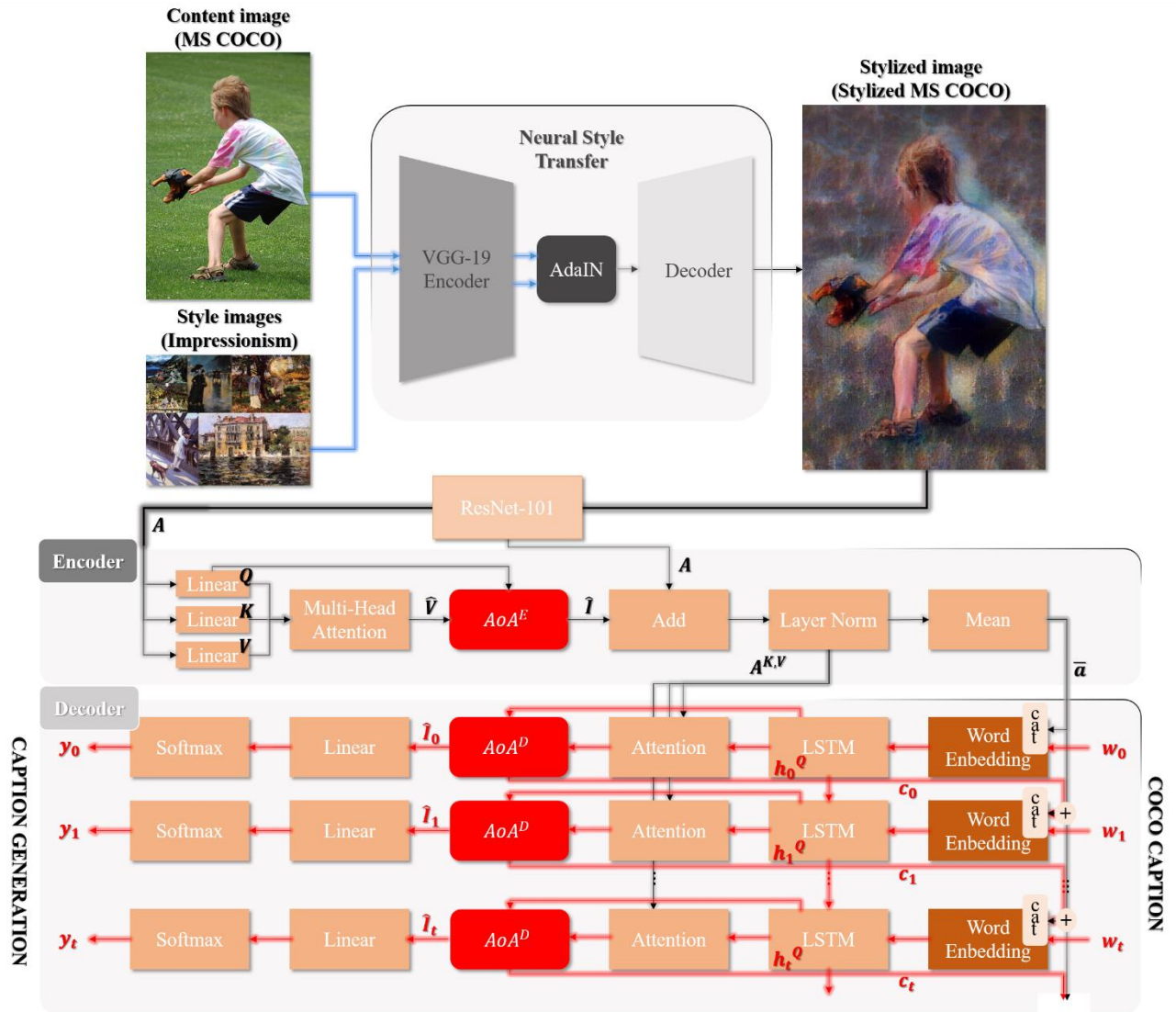


Figure 2. Deep learning network architecture for image caption generation from stylized

sections 2.3 and 2.4 respectively. The neural style transfer was implemented on MS-COCO dataset content images by providing impressionism style images [9]. The structure and hyper-parameters of these neural networks were adopted and implemented in the originally proposed manner to preserve the strengths and maintain reproducibility.

3.1 Network Architecture

Essentials of encoder and decoder architecture in deep learning is a span (or hull) of latent vector space. From the latent space, proper latent vectors are learned for loss function by training artificial neural networks. Our baseline models for style transfer and image captioning adopt the encoder-decoder architecture, as in Fig 2. In style transfer, AdaIN helps decoder enabling to present the latent vectors in the form of a stylized image. However, latent vectors in image captioning are considered as the meaning of local contexts. The attention mechanism performs the formation of global context vectors, and the AoA module makes it work seamlessly. The AoA module ultimately establishes a definitive attention score, which means the relationship among feature vectors of objects in the image. The AoA models were trained for 20 epochs on impressionism stylized images as well as on the original MS-COCO dataset separately, while other parameters and networks were preserved as per [17].

3.2 Evaluation Metrics

A sentence or expression of natural language can usually be described in multiple ways in the same language, while all the descriptions might be correct. Hence conventional measures for performance cannot be used to evaluate for this task. Therefore, we have employed different metrics, including BLEU [19], METEOR [20], ROUGE-L [21], CIDEr-D [22], and SPICE [23] in order to evaluate the proposed method. Bilingual Evaluation Understudy (BLEU) is a quantitative index or score to compare AI-generated image caption to ground-truth reference captions. This approach considers the occurrence of common n-grams in both generated and reference captions, regardless of the n-gram word order. BLEU score results in 100 as a perfect match and 0 as a perfect mismatch, but 30 to 40 score is generally considered as understandable to good correlation. N-grams of 1 through 4 words have been evaluated for this study. Metric for Evaluation of Translation with Explicit Ordering (METEOR) uses a weighted F-score instead of precision and recall, and incorrect word order is penalized, unlike the BLEU score. Thus, it has a better human judgment correlation. ROUGE incorporates language summary evaluation, and its variant based on the longest common subsequence (ROUGE-L) has been evaluated here. Another evaluation metric named CIDEr-D is commonly used with MS-COCO evaluations, an objective comparison for consensus-based evaluation based on human-likeness. Lastly, Semantic Propositional Image Caption Evaluation (SPICE) is a measure of an object, attribute and their relations efficient recovery from image captions.

4. EVALUATION

We evaluate MS-COCO, and stylized MS-COCO test data split on the network trained on stylized MS-COCO. In the same way, the evaluation is used on MS-COCO trained network for comparison purposes. The captions generated on all un-seen four test data split for both trained networks are analyzed using the evaluation metrics mentioned in the preceding section, which are arranged in Table 1.

As it is evident from the table’s right bottom that the impressionism imitated artwork generated captions from stylized MS-COCO trained network exhibits BLEU-4 score with four training captions of 30, which is rated as “understandable”. While BLEU 1 through BLUE-3 reflects even higher ratings, the best performance of the model for all evaluation metrics is achieved with four training captions. The image captions generated for the original MS-COCO test split on the same trained network score lower for all evaluation metrics, which is intuitive because it was trained only on stylized MS-COCO test data for this case. The performance of the deep network trained on original MS-COCO training data split with the same set of training hyper-parameters is also provided for baseline analysis. The captions generated from this network also perform better for stylized MS-COCO test set evaluation than the original MS-COCO test data scores. Moreover, among both MS-COCO test split evaluations, it performs better on a stylized MS-COCO trained network. It can be hypothesized that it is easier to learn abstract features from imitated artwork images than learning rich latent features from real-world images, especially when the training is restricted to fewer epochs.

Table 1: Evaluation of generated captions from Impressionism images

Performance			Trained Network									
			MS-COCO					Stylized MS-COCO				
No. of captions			1	2	3	4	5	1	2	3	4	5
Test set performance	MS-COCO	BLEU-1	59.96	57.91	61.81	60.5	54.93	47.7	59.83	61.13	61.64	60.31
		BLEU-2	41.28	39.14	43.15	41.63	35.95	28.16	41.09	42.45	42.9	41.55
		BLEU-3	28.5	26.82	30.02	28.74	24.2	17.58	28.23	29.49	29.96	28.95
		BLEU-4	20.16	18.93	21.25	20.19	16.62	11.61	19.92	20.93	21.37	20.6
		METEOR	18.36	17.9	19.3	18.87	16.98	13.37	18.46	19.03	19.36	18.91
		ROUGE-L	44.34	43.17	45.14	44.55	41.63	36.94	44.1	44.84	45.23	44.7
		CIDEr-D	53.55	50.97	60.95	58.28	44.74	22.88	56.16	59.71	60.97	58.98
		SPICE	11.3	10.8	12.41	11.91	10.07	5.88	11.44	11.96	12.42	11.96
	Stylized MS-COCO	BLEU-1	63.23	62.45	66.21	65.48	61.34	55.25	69.33	70.28	70.73	69.76
		BLEU-2	44.92	43.95	47.66	46.94	42.37	35.69	51.65	52.84	53.27	52.11
		BLEU-3	31.51	30.82	33.84	33.33	29.12	23.49	37.89	39.09	39.52	38.54
		BLEU-4	22.38	21.98	24.08	23.92	20.24	16.16	27.92	28.88	29.31	28.63
		METEOR	20.1	20.05	21.27	21.28	19.57	16.29	23.41	23.76	24.1	23.96
		ROUGE-L	46.55	45.94	47.96	47.67	45.21	41.02	50.88	51.5	51.75	51.34
		CIDEr-D	64.93	64.24	73.94	73.01	60.5	38.71	86.17	88.82	90.52	89.1
		SPICE	13.14	12.99	14.9	14.57	12.86	9.02	16.58	16.85	17.28	17.1

5. CONCLUSION

People with visual impairments cannot experience the visual artwork by the sense of vision. However, other senses can be utilized for their art appreciation. We have proposed a novel system that automatically generated a short description of visual artwork. When conveyed through the auditory sense, these descriptions can help PVIs decide the selection of artwork for further exploration. It can also escalate artwork cognizance by providing objective information and sensory actuation by artwork tactile representation. The proposed method extracts the objects and content from impressionism artwork and translates it to natural language captions using Attention on Attention artificial intelligence network. The network is trained on an unprecedentedly large number of impressionism art genre stylized images, and the quality of generated captions from test images is extensively evaluated. Experimental results have confirmed that the adopted AoA module applied to the language model has broad generality over learned features for numerous objects. Comparisons for the model's generalization capability are also provided by covering relevant metrics. This developed a non-visual information delivery multi-modal scheme that can minimize the information gap regarding visually impaired people's visual artwork appreciation.

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